



Could you cope with just a handful of sugar and one egg a week?

【BBC精选：英国二战饮食实验：每周1个鸡蛋、限量糖，竟比现代人更健康？】

Summary: During World War II, Britain faced potential food shortages due to German U-boat attacks on imports. Nutritionists Elsie Widdowson and Robert McCance designed a minimal diet using only UK-produced foods to test if health could be maintained under strict rationing. Their experiment, involving rigorous physical activity, proved that the limited diet—though bland—could sustain fitness. Their findings influenced government policy, including bread fortification with calcium to prevent deficiencies. Rationing lasted until 1954, but their legacy continues in nutrition science and support for female researchers.

摘要： 二战期间，德国潜艇威胁英国食品进口，导致物资短缺。营养学家埃尔西·威多森和罗伯特·麦坎斯设计了一项仅使用英国本土食物的极简饮食实验，测试严格配给制度下的健康维持情况。他们通过高强度体力活动证明，这种单调但均衡的饮食能够维持健康。其研究结果影响了政府决策，包括在面包中添加碳酸钙以预防营养不良。食品配给制持续至1954年，但他们的科学遗产仍持续影响着现代营养学研究和女性科学家支持计划。



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🎵 If you're watching this whilst tucking into a lunchtime sandwich

如果你正在#同时#吃着一份午餐三明治来看这个节目

🎵 made with avocados from Peru and prawns from Thailand,

三明治里夹着秘鲁牛油果和泰国虾仁,

🎵 it's probably hard to imagine a scenario where Britain could only eat food grown in the country.

你大概很难想象一种#情景#：英国只能吃自己种的食物。

▶ On top of that, you were limited to one egg a week and just a handful of sugar.

除此之外，你每周只能吃一个鸡蛋，只有一#把#糖。

▶ But that's the diet that was tested by two nutritionists in a unique experiment.

但这就是两位营养学家在一项独特实验中测试的饮食。

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▶ This experiment was devised during the early days of the Second World War.

这个实验是在二战初期设计的。

▶ and arguably it was healthier than the average diet today.

可以说，它比我们今天的平均饮食更健康。

▶ In the late 1930s Britain imported 70% of its food from overseas.

在20世纪30年代末，英国70%的食物依赖#海外#进口。

▶ It often relied on shipping routes, vulnerable to German U-boat attack.

这些运输航线常常#脆弱#地暴露在德国潜艇的攻击下。

▶ By January 1940, supplies were so scarce that food rationing was introduced.

到1940年1月，供应如此#稀缺#，以至于政府不得不推行食物配给制。

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▶ But the government was fearful that the country would be completely cut off by the U-boats.

但政府#担心#全国会被潜艇完全切断供应。

▶ They would have to keep its population fed and healthy.

他们必须让全国人民有饭吃，还能保持健康。

▶ They would use only food produced in the UK.

他们设想只使用英国本土生产的食物。

⏮ Before the war, nutritionists, Elsie Widerson and Robert MacKance had developed a set of food tables.

战前，营养学家埃尔西·威德森和罗伯特·麦肯斯制定了一套食物表格。

⏮ These tables looked at the nutritional value of meat, fish, fruit, bread, sweets and dairy.

这些表格分析了肉类、鱼类、水果、面包、糖果和#乳制品#的营养价值。

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⏮ They were no strangers to self-experimentation, even going so far as injection themselves with iron and strontium lactate,

他们对自我#实验#并不陌生。他们甚至给自己#注射#铁和#锶#乳酸盐。

⏮ to test how they were processed by the body.

他们这样做是为了测试这些物质在体内如何代谢。

⏮ These acts left them rolling around the floor in misery.

结果让他们痛苦得在地板上#惨叫翻滚#。

⏮ Together, they calculated the amount of food that would be available.

他们一起#计算#了如果德国切断英国港口时可获得的食物量。

⏮ This would be if Germany was able to cut off British ports.

这是在德国能封锁英国港口的情况下。

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⏮ They then devised it into a weekly allowance.

然后他们把这些食物设计成每周的#配额#。

⏮ The rations were meager to say the least.

说句实话，这些口粮相当#微薄#。

⏮ As well as that solitary egg and 140 grams of sugar a week,

除了那一个#孤零零#的鸡蛋和每周140克糖，

🎧 you could only have 450 grams of meat and fish combined.

你每周的肉和鱼加起来也只有450克。

🎧 113 grams of fats, the same with cheese, 170 grams of fruit,

113克脂肪、同样重量的奶酪，170克水果，

🎧 that's around two apples and a quarter of a pint of milk a day.

大约相当于两个苹果和每天四分之一品脱牛奶。

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🎧 British Pro Vegetables and Bread weren't restricted.

英国自产的蔬菜和面包没有被限制。

🎧 But the rationed foods represented less than half

但被配给的食物不到平常的一半，

🎧 and in some cases less than a third of the average weekly consumption.

有时甚至不到平常每周#消耗#量的三分之一。

🎧 This was before the war began.

这还是战争开始之前。

🎧 It was enough to survive but would it be enough to maintain fitness in the long term?

它足够维持生存，但能否长期保持#健康#呢？

🎧 Widerson and MacKance put their bodies on the line to find out.

威德森和麦肯斯用自己的身体去验证答案。

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🎧 In January 1940, they headed to the lake districts.

1940年1月，他们去了湖区。

▶ There they tested the diet whilst on a rigorous regime of cycling and mountain climbing.

他们在严格的骑行和登山#计划#中#同时#测试了这种饮食。

▶ On their return, they delivered a report to the government.

回去后，他们向政府提交了一份报告。

▶ The report confirmed that health and fitness could be maintained under this severe diet.

报告确认，在这种严格饮食下，人的健康和#体能#可以维持。

▶ They did, however, note a remarkable increase in flutchements.

不过，他们确实注意到胀气的显著增加。

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▶ The extreme diet, Widerson and MacKance tested wasn't actually needed in the end.

最终，威德森和麦肯斯测试的极端饮食并没有真正用上。

▶ This is because sufficient imports from the US managed to get to Britain throughout the war.

因为整个战争期间，来自美国的进口供应仍能到达英国。

▶ But it proved that the population would be able to manage on it.

但这证明了全国人民是可以靠它勉强度日的。

▶ Their findings were a key influence behind the government's decision.

他们的研究结果对政府的决定产生了关键影响。

▶ The decision was to leave bread and vegetables unrashing.

决定是不对面包和蔬菜实行配给。

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▶ Happy days if you liked potato sandwiches.

如果你喜欢土豆三明治，那真是快乐时光。

▶ The rationing that was introduced to lived around 3,000 calories a person a day.

实行的配给制大约能提供每人每天3000卡路里。

▶ Some manual labourers received extra.

一些#体力#劳动者还会得到额外补充。

▶ Nevertheless, the diet was often bland.

然而，这种饮食通常相当#寡淡#。

▶ Thanks to the Ministry of Food,

多亏了食品部，

▶ households were inspired to create such culinary delights as a walton pie.

人们被鼓励创造出诸如沃尔顿#馅饼#之类的#烹饪#创意。

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▶ This was a mix of sweet, turnip and cauliflower covered with short-crust pastry or mashed potato.

它是由甜菜、萝卜和花椰菜混合，再盖上#酥皮#或土豆泥制成的。

▶ It's fair to say it wasn't much loved.

可以坦率地说，它并不受欢迎。

▶ Blandness wasn't the only issue.

平淡无味不是唯一的问题。

▶ In further studies, Widerson and MacKance realised that people wouldn't get as much calcium as they needed.

在进一步研究中，威德森和麦肯斯发现人们摄入的#钙#远远不足。

▶ This would be if milk and cheese were to be severely rationed.

这是因为牛奶和奶酪被严格配给。

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▶ This might prove disastrous.

这可能会导致#灾难性#的结果。

▶ A lack of calcium can lead to the bone disease, wickets.

缺乏#钙#会导致骨病——佝偻病。

▶ In 1941, based on their advice, the government brought in the fortification of bread with calcium.

1941年，根据他们的建议，政府开始给面包添加#钙#。

▶ This was in the form of powdered chalk.

这是通过粉状#石灰粉#来实现的。

▶ This ensured people got enough of the mineral in their diet.

这保证了人们在饮食中摄入足够的这种矿物质。

▶ This was despite the restrictions on dairy.

尽管对#乳制品#有限制。

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▶ Food rationing lasted until 1954.

食物配给一直持续到1954年。

▶ But even after ended, bread continued to be fortified with calcium.

但即使在结束后，面包仍然被添加#钙#。

▶ It's still the case in the UK today.

在英国至今依然如此。

▶ If you're wondering who to thank for your healthy bones.

如果你想知道要感谢谁才能让你有健康的骨骼，

▶ Whilst rationing wasn't easy for Britain, it got the nation through a wartime crisis in pretty good shape.

虽然配给制对英国并不轻松，但它让全国在战时危机中保持了相当好的状态。

▶ and arguably even MacKance's and Widerson's extreme diet was healthier than our average diet today.

甚至可以说，麦肯斯和威德森的极端饮食比我们今天的平均饮食还要健康。

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▶ Our diet is so dominated by high fat, high sugar and high salt foods.

我们现在的饮食过于依赖高脂肪、高糖和高盐的食物。

▶ The pair continued to demonstrate that they were prepared to go the extra mile in the name of science.

这两位继续证明，他们愿意为了科学付出额外的努力。

▶ Widerson famously found a dead baby seal on a beach in Scotland.

威德森曾经著名地在苏格兰的海滩上发现了一只死去的小海豹。

▶ She drove it back to Cambridge in the boot of her car.

她把它#开车#带回了剑桥，放在汽车后备箱里。

▶ She did this in order to further her research into body fat composition.

她这样做是为了进一步研究身体脂肪的#组成#。

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▶ Widersons and MacKance's legacies live on.

威德森和麦肯斯的遗产至今犹存。

▶ This is not just in terms of nutrition.

这不仅仅体现在#营养学#方面。

▶ When she died, Widerson left money to enable female scientists to bounce their research with family life.

当她去世时，她留下资金，帮助女性科学家在科研与家庭生活之间#兼顾#。

▶ This supports the next generation of pioneering women.

这支持了新一代开拓性的女性学者。

▶ They will help solve our complex scientific problems today.

她们将帮助解决当今复杂的科学问题。

